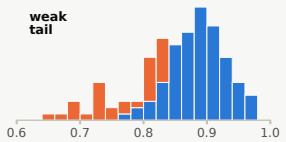
User-Bound Limits of Calibration-Free RSVP-EEG Decoding: Evidence From Eleven Decoders and 122 Participants

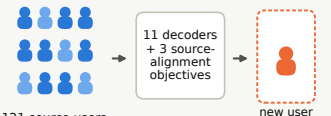
PROBLEM

Calibration-free RSVP brain-computer interfaces serve a new user without a recording session, but accuracy varies widely between users.

Is the difficulty set by the decoder, or by the user?



METHODS

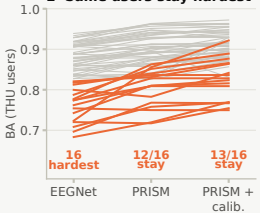


- 3 public RSVP datasets, 122 participants
- xDAWN-RG + 10 deep nets; CORAL/MMD/DANN
- Strict leave-one-subject-out evaluation
- Calibration on each user's own data
- ERP descriptors from disjoint trials

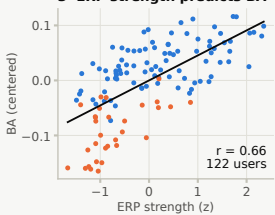
FINDINGS

orange = hardest quartile of users

2 Same users stay hardest



3 ERP strength predicts BA



1 Participants explain 77–94% of BA variance; decoders only 2–11%

2 Better decoders, alignment, and calibration leave the same users hardest

3 Weak ERP strength explains the difficulty (r = 0.66, disjoint trials)

CONCLUSION

Difficulty in calibration-free RSVP-EEG decoding is bound to the user: users with weak ERPs stay hardest whatever the decoder, alignment, or calibration. New users should be screened with about 40 target trials (2–7 min), and studies should report user-level accuracy.